

Social Touch as an Allostatic Tool in Cooperative Multi-Agent Reinforcement Learning

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Abstract

This study explores the computational modeling of social touch in a cooperative multi-agent reinforcement learning (MARL) setting. Inspired by research on social allostasis and affective neurobiology, we implement a novel model of social touch in artificial agents within a cooperative game setting, where interoception-inspired internal states are modulated by eating and agent-to-agent contact. A memory-augmented Proximal Policy Optimization (PPO) is used to explore and exploit social touch for cooperative reward optimization and internal state regulation. Our paradigm reveals that agents trained with modulatory social touch achieve better game performance, even in environments where touch is ineffective. We observe an evolution of touch behavior from frequent exploration to selective usage, showing better internal-state regulation and adaptability to heterogeneous starting conditions between agents. The findings demonstrate how interactive systems might incorporate non-verbal communication cues to enhance cooperation, with implications for designing more intuitive human-AI and robot-robot collaborative systems.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Multi-agent reinforcement learning*.

Keywords

Cooperation, Multi-agent Reinforcement Learning, Social Allostasis, Affective Neuroscience, Embodied Cognition, Social Touch, Haptics, HCI

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1 Introduction

Artificial multi-agent systems are an exciting intersection of robotics, biology, and social sciences, offering unique insights into the emergence and evolution of complex behaviors. Using reinforcement learning to simulate environments where agents learn through trial and error, we can explore how social behaviors such as collaboration and cooperation can evolve and stabilize in artificial systems. This approach represents a symbiosis between interaction technology and social and cognitive sciences and has the potential to both expand our understanding of social system dynamics and to improve the performance and integration of interaction models in real-life deployment.

One particularly intriguing area to study at this intersection of social behavior and artificial systems is allostasis. Allostasis refers to the active process by which organisms maintain physiological stability through change. Allostasis allows for the use of external resources, including social resources, for adaptive changes in the organism in response to environmental demands and anticipated needs. Social allostasis, which refers to the way individuals use social interactions to maintain physiological balance, provides a bridging framework to model how artificial agents can cooperate for their own individual and collective goals [9, 16].

This work explores social touch as a key mechanism of social allostasis in multi-agent artificial systems. Social touch is described as an intentional contact between two humans with primary social and emotional goals and is recognized as an allostatic tool [25], with its role in promoting cooperative behavior well-established in human interaction research [5, 12]. In this work, we explore whether social allostasis mediated by social touch improves agents' performance in systems that rely on cooperation by directly answering the following research questions:

- **R1:** Does social touch behavior (and specifically, touch behavior aimed at allostatic regulation) spontaneously emerge from the strategies adopted by agents to maximize rewards?
- **R2:** Does the use of social touch correlate with game performance?
- **R3:** Do agents trained with touch-based regulation maintain performance advantages even when deployed in environments where touch has no effect on internal state regulation?

To answer these questions, we customized a multi-agent benchmark environment with novel agents that can use social touch to modulate their internal states. Our agent model simulates arousal and satiety as key internal states, with arousal directly affecting an agent's perceptual range and satiety gating the eating behavior. We then compare agents capable of modulating these states through social touch with those who cannot and we show that:

- Touch-enabled agents achieve significantly higher cumulative rewards and show more stable internal states.
- The observed touch behavior between agents evolves from frequent interactions to more selective and strategic usage over time.
- Even when evaluated in environments where touch has no impact on agents, agents trained leveraging touch-based allostasis outperform their counterparts in terms of rewards obtained, total number of units of food eaten, higher satiety and lower arousal.
- In scenarios where agents have different starting internal resources (internal states of arousal and satiety), touch-enabled systems show an equalizing effect, reducing performance disparities between agents.

This paper details our system architecture, presents these findings, and discusses implications for emergent behavior and human-computer interaction design.

2 Background and Related Work

Cooperation forms a fundamental building block of society, studied across multiple disciplines [3, 4, 22, 23]. In multi-agent reinforcement learning (MARL), key challenges remain in understanding how individual agents' capabilities support coordination and cooperation in group settings [2, 8]. Researchers note that multi-agent scenarios where agents can have differences in objectives require studying effective communication strategies [3], especially in tasks with sparse rewards [24]. Recent work highlights the need for lightweight, biologically-inspired communication mechanisms in multi-agent systems [10], as researchers increasingly look to social and biological systems for efficient alternatives [8]. Natural systems demonstrate how simple signaling systems can achieve sophisticated coordination, offering promising directions for artificial design [3, 8].

2.1 Allostasis in MARL Systems

The study of allostasis has gained significant traction in recent years [15]. Computational models have primarily focused on exploring the advantages of the predictive component of allostasis [14]. Homeostatic Reinforcement Learning (HRL), introduced by Keramati and Gutkin [11], represents a significant refinement of traditional reinforcement learning theories, grounding learning protocols on the individual's internal state. However, to our knowledge, no prior work has integrated allostasis specifically in multi-agent reinforcement learning. Our study aims to extend the concept of allostasis to MARL, with an added context-relevant social component of game-relevant affect.

2.2 Haptics to Bridge Cooperation and Allostasis

Research in Human-Computer Interaction and Affective Computing recognizes that haptic interfaces have significant potential to promote prosocial behavior. Studies indicate that haptic feedback can reduce feelings of separation between people [21] and encourages cooperation even between ingroup and outgroup members [18]. In biology research, touch is recognized as an important tool for allostatic regulation in biological systems. Studies demonstrate

that touch contributes to the predictive regulation of both short and long-term physiological resources [7], primarily by modulating arousal states through interoceptive pathways that signal bodily cues [1, 20]. These biological mechanisms, combined with advances in haptics research, suggest promising applications for implementing regulatory systems in artificial agents through simulated touch interactions, particularly in tasks requiring attention modulation and resource management.

3 Methods

3.1 Environment

This study customizes the PettingZoo Waterworld environment originally developed by Stanford Intelligent Systems Laboratory (SISL) [19]. The environment simulates a 2D aquatic environment $([0,1] \times [0,1])$ where agents ("pursuers") must cooperatively catch "evaders," which serve as food resources, while avoiding poison. Cooperation is enforced by requiring multiple agents' proximity for food consumption, where proximity is detected as a collision by the physics simulation. We augmented the agents to have an internal representation of satiety and arousal states, in line with the objectives of the game, and to have arousal modulated by agent-to-agent contact in the environment (agent interpersonal touch). It is important to note that social touch is not explicitly designed or rewarded in agents. We only considered 2 agents, 5 food sources, and 10 poison units at a time.

Table 1: Environment Configuration

Component	Value	Function
Agents	2	Pursuers in 2D space
Food Sources	5	Nutrition value 0.1-0.5
Poison Units	10	Negative reward on contact

3.2 Agent Architecture

Pursuers sense their environment through 30 equidistant rays detecting nearby entities. Sensor range adjusts dynamically with arousal levels.

3.2.1 Internal States. We chose arousal and satiety as internal states to allostatically regulate. Research indicates that arousal levels correlate with attentional capacity [26], with arousing stimuli improving focus and response times [13]. Arousal is particularly well-suited for allostatic regulation through social touch, as demonstrated by extensive research showing touch's direct modulatory effects on arousal through interoceptive pathways [1, 6, 20]. In biological systems, satiety provides direct physiological links to energy management [17], working alongside arousal for regulatory control. Increased hunger cues can heighten arousal, while higher arousal levels can, in turn, induce further hunger.

In our system, arousal (physiological activation, $[-1,1]$) modulates sensor range and is affected by social contact and hunger. Satiety (energy level, $[0-5.0]$) increases with eating and decreases over time.

3.2.2 Social Touch. Agent-to-agent contact is detected by collision handlers and triggers arousal modulation through configurable

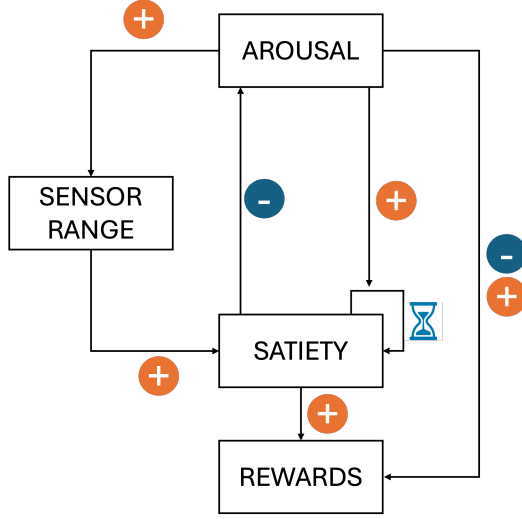


Figure 1: Relationship between internal parameters influenced by our modifications to the pettingZoo environment. ‘+’ symbols represent positive effects between parameters, ‘-’ symbols indicate negative effects. The direction of the arrow indicate causation. The diagram shows how Arousal affects sensor range facilitating eating behaviors ultimately increasing satiety. Satiety is inversely related to arousal (which increases with hunger) and is directly proportional to the agents’ achieved rewards. Furthermore, satiety decreases over time, but the rate of decay decreases for low levels of arousal and increases for high levels. Finally, extreme levels of arousals (both too high and too low) are directly penalized (hence the double symbol connecting ‘Arousal’ and ‘Rewards’).

modes: "average" (mean arousal of two agents), "cooperative" (maximum), "competitive" (minimum), or no_effect (disabled).

The impact of social touch is implemented through modulation of arousal levels, which in turn affects agent perception and resource management. Arousal affects both sensor range ($\text{sensor_range} = \text{base_range} \cdot (1 + \text{arousal})$) and satiety decay ($\text{satiety_decay} = \text{base_decay} \cdot (1 + \text{arousal})$). This creates an implicit pressure to regulate arousal through social touch: high arousal increases perceptual range but depletes energy faster.

Table 2: Reward Components

Component	Value
Food Consumption	10
Food Encounter	0.01
Thrust Penalty	$-0.5 \times \text{acceleration} $
Poison Contact	-1
Satiety Bonus	$5 \times \text{current satiety}$
Arousal Penalty	$-2 \times \text{current arousal} $

3.2.3 Rewards.

3.3 Learning Architecture

We implement a memory-augmented PPO with Long Short-Term Memory (LSTM) to capture temporal dependencies, using centralized learning and decentralized execution in an actor-critic structure. Training uses vectorized environments and linear learning rate scheduling.

3.4 Training and Evaluation

Training proceeded for up to 10M steps with early stopping based on evaluation performance every 10K steps. We evaluated agents across three conditions: (1) native training environments, (2) environments without touch modulation, and (3) heterogeneous initial states (arousal/satiety: 0.1 vs 0.9). Each evaluation comprised 20 games of 500 frames, measuring rewards, internal states, and touch behaviors.

4 Results

Our analysis reveals how social touch can serve as an effective regulatory mechanism in multi-agent systems, with implications for designing social touch interfaces. We evaluate two conditions: agents trained in environments where social touch modulates internal states (“touch-trained”) and those where touch occurs without modulation (“non-modulatory”).

4.1 Training Outcomes

We compare the performance and behavior of agents in both environments to compare the dynamics of social touch and its impact on agent performance developing over training steps until no performance improvement.

In the “touch-trained” modulatory condition, agents initially explored social touch more frequently but reduced their usage in later stages, suggesting a shift towards more selective and efficient utilization. This adaptation coincided with the start of a gradual increase in rolling rewards despite fewer eating events. In contrast, non-modulatory agents showed inconsistent reward progression even with more frequent eating, suggesting touch-based arousal modulation enhanced reward accumulation.

The modulatory condition also showed improved internal state management in the later stages of training, with satiety levels increasing despite reduced eating, and a decrease in arousal, which potentially indicate group-level management of internal states. We recognize longer training could further stabilize these behaviors.

4.2 Performance Analysis

Non-parametric analysis between game performances revealed significant differences ($p < 0.05$) across all measured variables. Within their respective environments, touch-enabled agents significantly outperformed in cumulative rewards ($U = 304.0$, $p = 0.0051$, median $df1$: 6376.98, $df2$: 4395.59), showed better satiety maintenance ($U = 303.0$, $p = 0.0053$), and demonstrated more effective arousal regulation ($U = 126.0$, $p = 0.0456$). The highly significant difference in social touch frequency ($p < 0.0001$) indicates that agents learned to use touch more selectively and strategically during game performance.

Notably, these benefits persisted even when agents were tested in transfer environments where touch had no modulatory effect.

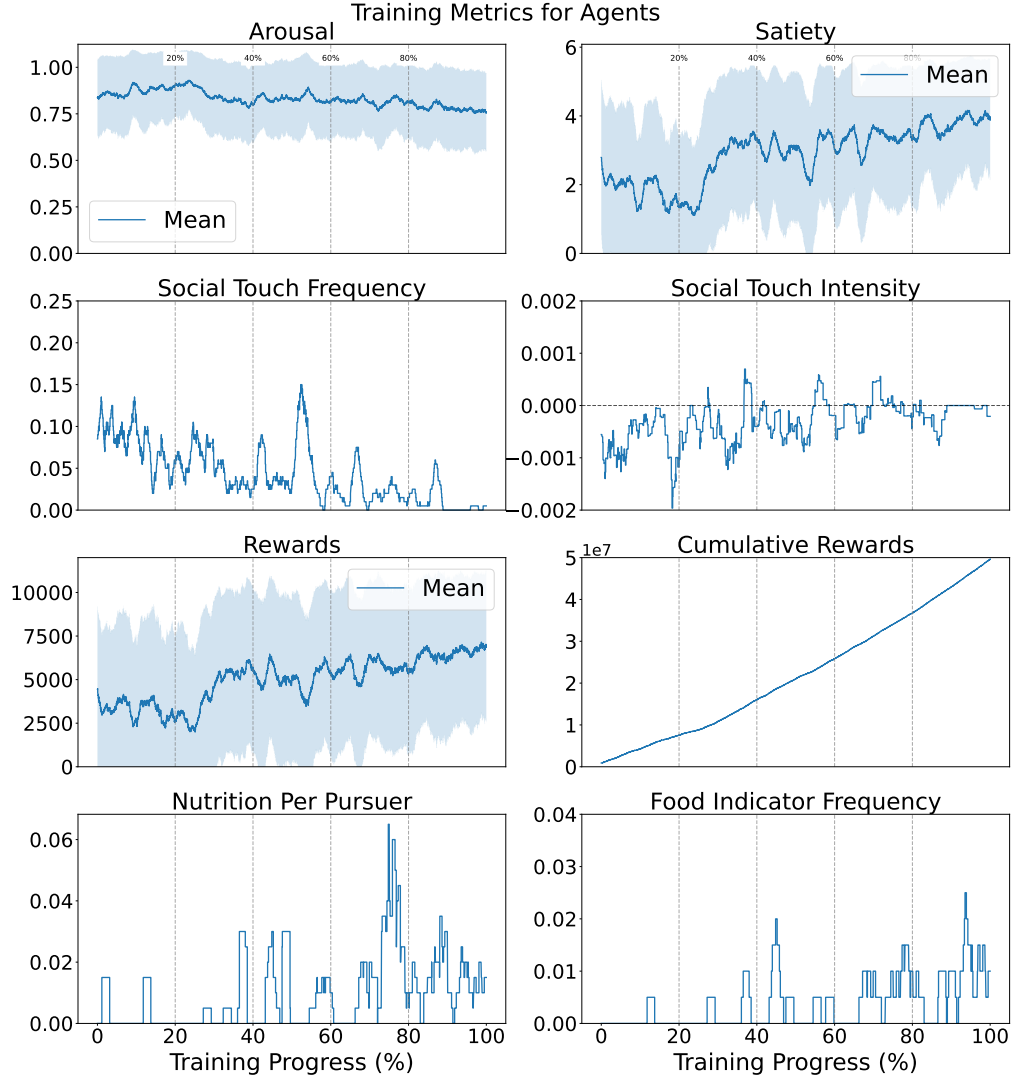


Figure 2: Modulatory Social Touch Environment Metrics of Interest observed during Training (up to 10M steps, terminated early). Calculated on a rolling window of 1% of the total number of time points in the dataset for consistent smoothing. Frequency of metrics for the rolling window is plotted in x against the rolling step timeline on y axis. Rather than absolute values, progression of values is inspected. Rolling rewards is average reward per episode and cumulative rewards is the sum of all rewards obtained so far. Evader eaten frequency is the frequency at which food is eaten. Food indicator frequency refers to instances where an agent comes into contact with food but cannot eat due to absence of the other agent, but receives a small shaping reward to encourage this action. Due to the centralized learning policy, we plot these trends for one agent, which can be treated as one multi-jointed agent made up of two “Pursuers”. Note the increase in rolling rewards and concurrent increase in satiety and decrease in arousal starting from 50000 rolling steps.

Touch-trained agents maintained better performance in cumulative rewards and satiety levels ($p = 0.0764$ and $p = 0.1019$ respectively), and used social touch significantly more sparingly presumably as strategic ($U = 6.0$, $p < 0.0001$). The difference in cooperative food consumption remained statistically significant ($U = 263.0$, $p = 0.0415$), suggesting that cooperative behaviors developed during touch-modulation training persisted even in non-modulatory environments.

4.3 Adaptability to Heterogeneous Conditions

Touch-trained agents effectively used social touch to regulate disparate internal states. To investigate the potential benefit of social touch-enabled cooperation in unfair group dynamics—i.e., different internal states/need for food—we also evaluated agents with different initial arousal and satiety levels (0.9 vs 0.1). In evaluations with unequal initial conditions, touch-trained agents maintained

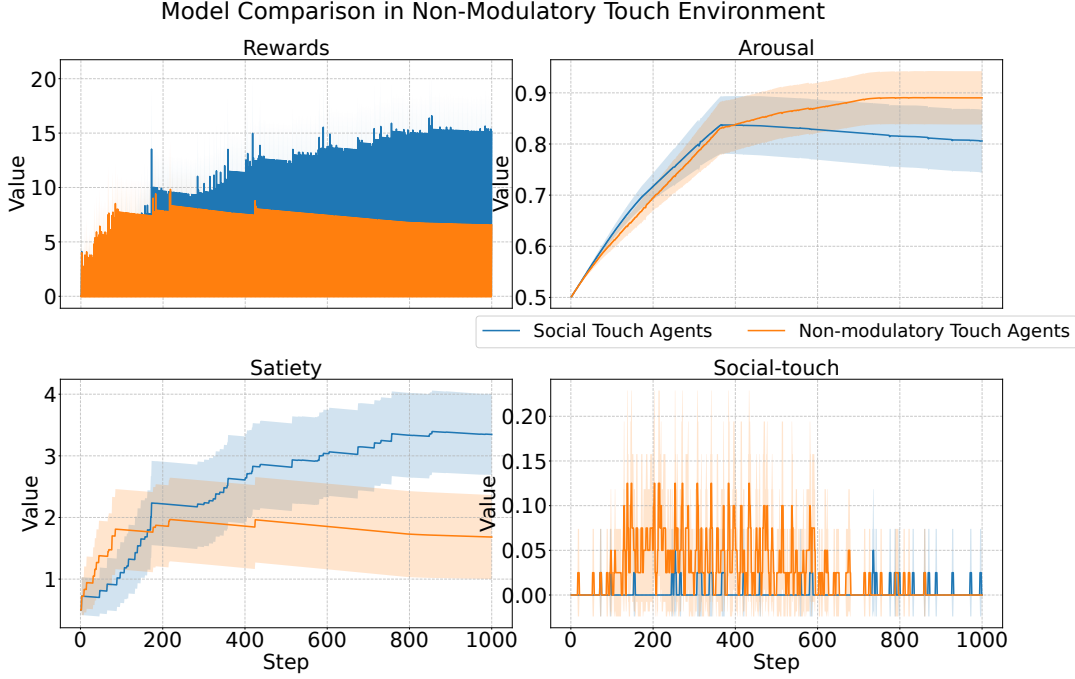


Figure 3: Training Metrics for Agents. Calculated on a rolling window of 1% of the total number of time points in the dataset for consistent smoothing. The plots show arousal (top left), satiety (top right), social touch frequency (bottom left), and social touch intensity (bottom right) over training progress. Note the decrease in arousal and increase in satiety around 50% of training, coinciding with more strategic usage of social touch.

performance advantages over non-modulatory agents in cumulative rewards ($U = 125.0$, $p = 0.0439$), arousal regulation ($U = 291.0$, $p = 0.0138$), and task completion ($U = 118.0$, $p = 0.0150$).

Interestingly, while touch frequency remained consistent between homogeneous and heterogeneous conditions ($U = 200.0$, $p = 1.0$), touch modulation intensity varied significantly based on initial states ($U = 327.0$, $p = 0.0005$). This reveals that agents adapted to initial state disparities not by changing touch frequency but by adjusting modulation intensity—demonstrating sophisticated regulatory use of social touch.

5 Discussion

Our study demonstrates how social touch can serve as an effective regulatory mechanism in multi-agent systems. The prosocial effects of touch within groups and the relationship between arousal, perception, and performance are grounded in the literature on social allostasis [1] and neurobiology [20], providing a theoretical framework for designing touch-based social interactions in artificial systems.

5.1 Emergence of Social Touch and Regulatory Potential

We observed the emergence of strategic touch usage in our agents both while tracking training progression and during evaluation. Initially frequent but random touch interactions evolved into selective, context-appropriate usage in the final stages of training, coinciding

with improved performance and more stable internal states. At evaluation, agents trained in the modulatory touch condition showed more efficient performance compared to the non-modulatory condition across multiple performance metrics. A significant difference in social touch interactions supports our claim that these agents learned to use social touch strategically.

The emergence of selective touch coincided with improved internal state regulation. While we did not directly analyze the temporal dynamics of touch and allostasis, partly due to the limitations of our monitoring, our results suggest that agents discovered these regulatory benefits. Further analysis before and after touch events could clarify this relationship. The strategic use of touch, despite the absence of explicit rewards for contact, indicates its role in internal state regulation.

5.2 Touch and Performance Correlation

Nonparametric analysis revealed that touch-trained agents significantly outperformed non-modulatory agents across multiple metrics, even in game environments where touch provided no direct benefits. Here, agents can still engage in social touch by selectively choosing or avoiding physical proximity. Touch-trained agents achieved higher cumulative rewards ($U = 304.0$, $p = 0.0051$, median: 6376.98 vs 4395.59) and showed more effective task completion (evaders eaten: $U = 297.5$, $p = 0.0028$). Notably, these agents engaged in significantly fewer but more strategic touch interactions ($U = 17.5$, $p < 0.0001$), suggesting learned selective use of social touch, and correlating with game performance.

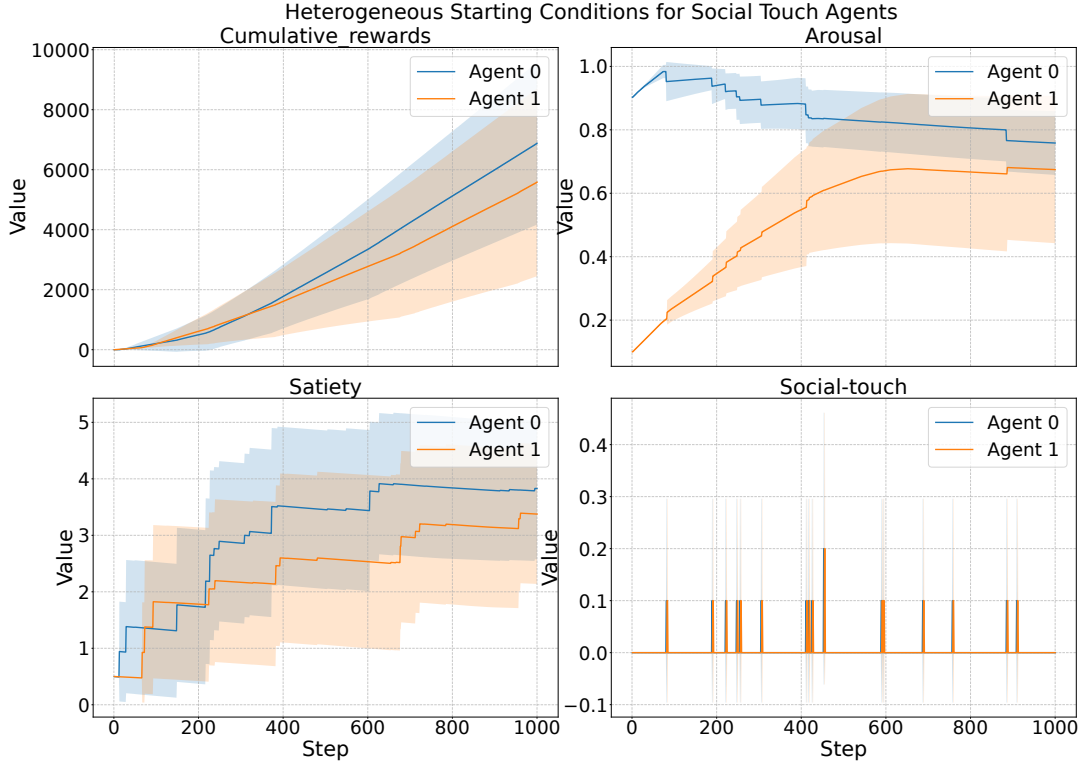


Figure 4: Performance metrics of touch-trained agents in Modulatory Social Touch Environment with different initial arousal states. The plots show cumulative rewards (top left), arousal (top right), satiety (bottom left), and social touch frequency (bottom right) over game steps. Each line represents a different agent, with shaded areas indicating 95% confidence intervals. Note the difference between arousal values at the start of the episodes, converging throughout the games.

The strategic advantage of touch training in game performance became particularly evident in heterogeneous starting conditions. Touch-trained agents showed superior performance in cumulative rewards ($U = 125.0, p = 0.0439$) and demonstrated more effective arousal regulation ($U = 291.0, p = 0.0138$). While touch frequency remained consistent between conditions ($U = 200.0, p = 1.0$), touch modulation intensity varied significantly ($U = 327.0, p = 0.0005$), indicating social touch allowed for sophisticated adaptation to initial state disparities, which arguably helped with game performance.

5.3 Generalization of Touch Benefits

The persistence of regulatory benefits in touch-trained agents, even in environments where touch cannot modulate allostasis, suggests that these agents learn adaptive cooperative behaviors that remain effective beyond their original training conditions. The generalization of learned behaviors aligns with social psychology research, which shows that touch-mediated cooperative behaviors persist across contexts. For HCI applications, this suggests that systems trained with social touch capabilities could maintain effective resource allocation and goal-sharing even when direct touch isn't possible. Such robust transfer of cooperative behaviors is particularly valuable when deploying autonomous systems in diverse environments with constrained communication channels.

6 Conclusion

This work demonstrates how biologically-inspired social touch can enhance cooperative behavior in artificial systems, with significant implications for HCI. By implementing a reductionist model of interoception and social touch in artificial agents, we show how simple, biologically-inspired interactions can lead to complex, adaptive behaviors in cooperative settings. Our findings suggest that incorporating simple contact-based regulatory mechanisms on key internal states can lead to more adaptive and effective interactions in scenarios where subtle, non-verbal communication plays a crucial role.

Our findings have direct implications for HCI applications. The successful integration of touch-based regulation suggests novel approaches to human-robot interaction, particularly in scenarios requiring goal attunement and resource management. Potential applications include therapeutic robots, elder care systems, and remote collaboration tools.

While our current two-agent system demonstrates clear benefits, future work should explore both scaling to larger populations with more complex social dynamics and extending this framework to more diverse cooperative game environments to confirm generalization. Second, comparing touch with alternative communication channels—such as explicit signaling when internal states reach critical thresholds—could clarify whether the benefits stem from

physical contact specifically or from any form of state communication. These investigations would help distinguish between the specific benefits of touch and the broader advantages of affect-based communication in cooperative scenarios.

This work bridges artificial systems and social psychology, offering empirical validation for implementing social touch in haptic interfaces. Our framework provides a foundation for integrating affective behaviors into HCI design, advancing the development of more socially adept artificial systems.

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We acknowledge the PettingZoo library <https://pettingzoo.farama.org/> as a basis of our work, which is supported by the open MIT license. The code for our customized implementation is available at <https://github.com/yidilozdemir/PettingZoo-Haptic-Waterworld/>.

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A Technical Implementation Details

The code from PettingZoo Waterworld environment <https://pettingzoo.farama.org/> was customized primarily in the ways outlined in the code provided below. Our modifications focused on implementing internal states (arousal and satiety), social touch mechanisms, and related reward structures. For the full codebase and detailed documentation, please refer to our open-source implementation at <https://github.com/yidilozdemir/PettingZoo-Haptic-Waterworld/>.

A.1 Agent Architecture

A.1.1 Physical Attributes.

- Size: Radius is reference point. Food has 2x radius, poisons 0.75x radius
- Position: (x, y) coordinate space in $[0, 1]$
- Movement: Velocity vector with max acceleration pursuer_max_accel
- Sensors: 30 equidistant rays detecting food, poison, pursuers
- Sensor Range: Dynamic based on arousal level

A.1.2 Internal States.

- Satiety: Energy level initialized at 0.5, max 5.0
- Hunger: Calculated as $(1 - \text{satiety})$ each timestep
- Arousal: Physiological activation in $[-1, 1]$, initialized at 0.5

A.1.3 Parameters.

- satiety_decay_rate: Satiety decrease rate
- arousal_decay_rate: Arousal baseline return rate

- haptic_modulation_type: Social touch influence type
- haptic_weight: Touch influence strength

Core update logic:

```
# Update arousal
hunger = self.max_satiety - self.satiety
self.arousal += (
    hunger * 0.01 * dt + self.social_haptic_modulation
)
self.arousal = np.clip(self.arousal, -1, 1)

# Update satiety
satiety_decrease_rate = self.satiety_decay_rate * (
    1 + self.arousal
)
self.satiety = max(self.satiety - satiety_decrease_rate * dt, 0)

# Decay arousal
if self.arousal > 0:
    self.arousal = max(
        0, self.arousal - self.arousal_decay_rate * dt
    )
else:
    self.arousal = min(
        0, self.arousal + self.arousal_decay_rate * dt
    )
self.arousal = np.clip(self.arousal, -1, 1)

# Update sensors
self.sensor_range = self.sensor_range * (1 + self.arousal)
```

A.2 Agent Actions

A.2.1 Collision Detection.

- add_handlers manages collision interactions
- pursuer_evader_begin_callback for eating events
- pursuer_pursuer_touch_callback for social touch

A.2.2 Eating Mechanism.

```
def eat(self, food_nutrition):
    self.satiety = min(1, self.satiety + food_nutrition)
```

A.2.3 Social Touch Types.

- average: Mean arousal of contacts
- cooperative: Maximum arousal
- competitive: Minimum arousal
- no_effect: Disabled modulation

A.3 Core Update Functions

A.3.1 Step Function Implementation.

```
def step(self, action, agent_id, is_last):
    # Scale action based on pursuer maximum acceleration
    action = np.asarray(action) * self.pursuer_max_accel
    action = action.reshape(2)
    thrust = np.linalg.norm(action)
    if thrust > self.pursuer_max_accel:
```

```
    action = action * (self.pursuer_max_accel / thrust)
```

```
p = self.pursuers[agent_id]
# Update pursuer velocity
_velocity = np.clip(
    p.body.velocity + action * self.pixel_scale,
    -p.max_speed,
    p.max_speed,
)
p.body.velocity = tuple(_velocity)
p.reset_velocity(_velocity[0], _velocity[1])
```

```
# Calculate thrust penalty (control reward)
accel_penalty = self.thrust_penalty * math.sqrt(
    (action**2).sum()
)
```

```
self.control_rewards = (
    (accel_penalty / self.n_pursuers)
    * np.ones(self.n_pursuers)
    * (1 - self.local_ratio)
)
self.control_rewards[agent_id] += (
    accel_penalty * self.local_ratio
)
```

```
if is_last:
```

```
    # Step physics simulation
    self.space.step(1 / self.FPS)
    obs_list = self.observe_list()
    self.last_obs = obs_list
```

```
# Update all pursuers' internal states
for pursuer in self.pursuers:
    pursuer.update_awareness(
        1 / self.FPS, self.pursuers, self.evaders
    )
```

```
# Reset behavior rewards
self.behavior_rewards = [
    0 for _ in range(self.n_pursuers)
]
nutrition_per_pursuer = 0
```

```
# Process food consumption events
for evader in self.evaders:
    pursuers_eating = [
        pursuer
        for pursuer in self.pursuers
        if pursuer.shape.food_indicator != 0
        and pursuer.satiety < pursuer.max_satiety
    ]
    if pursuers_eating:
        # Food is consumed
        nutrition_per_pursuer = (
            evader.nutrition / len(pursuers_eating)
        )
        for pursuer in pursuers_eating:
```

```

        pursuer.eat(nutrition_per_pursuer)
        pursuer_index = self.pursuers.index(
            pursuer
        )
        self.behavior_rewards[
            pursuer_index
        ] += int(self.food_reward)
        evader.eaten = True
    else:
        # Food is encountered but not consumed
        for pursuer in self.pursuers:
            if (
                pursuer.shape.food_touched_indicator
                > 0
            ):
                pursuer_index = self.pursuers.index(
                    pursuer
                )
                self.behavior_rewards[
                    pursuer_index
                ] += int(self.encounter_reward)

```

```

# Handle poison collisions
for poison in self.poisons:
    for pursuer in self.pursuers:
        if pursuer.shape.poison_indicator > 0:
            pursuer_index = self.pursuers.index(
                pursuer
            )
            self.behavior_rewards[
                pursuer_index
            ] += int(self.poison_reward)
            # Reset poison position
            x, y = self._generate_coord(
                poison.shape.radius
            )
            poison.body.position = x, y

# Calculate final rewards
self._calculate_rewards()

```